A shortest-odd-cycle reduction for biaffine extremality

Human Compression project: bounded L025 proof

Abstract

The fixed preprint's Theorem 4.3 uses a lower bound for bipartite competitors to make an unrestricted extremality assertion. We prove an elementary odd-cycle bound which settles every even prime-power degree at least eight. Together with the retained degree-four computation and odd-degree parity, this supplies the missing unrestricted extremality step for every prime power $q \geq 3$. This note proves the claimed result with its proper range; it does not declare the source theorem false.

1 A uniform elementary bound

All graphs are finite and simple. The odd girth of a nonbipartite graph is the length of its shortest odd cycle.

Lemma 1. *Let C be a shortest odd cycle, of length ℓ, and let a path of length r have distinct endpoints on C and all its internal vertices outside C. One of the two arcs of C between its endpoints has length at most r and parity equal to r.*

Proof. Exactly one of the two arcs has parity different from r. Its union with the path is a simple odd cycle, so its length is at least ℓ. The complementary arc consequently has length at most r. Since ℓ is odd, this complementary arc has the same parity as r. □

Proposition 2. *A nonbipartite k-regular graph of girth at least six and odd girth ℓ has*

$$|V| \;\geq\; \ell(k-1) + \left\lceil \frac{\ell(k-2)^2}{3} \right\rceil = \left\lceil \frac{\ell(k^2-k+1)}{3} \right\rceil . \tag{1}$$

In particular $\ell \geq 7$.

Proof. Index a shortest odd cycle by c_i, $i \in \mathbb{Z}/\ell\mathbb{Z}$. It has no chord: a chord and one of its two arcs would give a shorter odd cycle. No outside vertex can have two neighbors on C. Indeed, Lemma 1 for a path of length two supplies a two-edge arc, giving a quadrilateral. Thus the first shell S, consisting of all neighbors of C outside C, is the disjoint union of sets S_i, each of size $k-2$, with every element of S_i adjacent to c_i alone on the cycle.

Suppose $a \in S_i$ and $b \in S_j$ are adjacent. Equality $i = j$ would give a triangle. Otherwise the path c_i, a, b, c_j and Lemma 1 give an arc of odd length at most three. Length one would give a quadrilateral. Therefore $j - i = \pm 3$ modulo ℓ. A vertex $a \in S_i$ has at most one neighbor in either S_{i+3} or S_{i-3}: two in the same set would form a quadrilateral with that set's cycle vertex. In fact it cannot have one in each. If these were $x \in S_{i-3}$ and $z \in S_{i+3}$, the path $c_{i-3}, x, a, z, c_{i+3}$ would have length four and have all internal vertices outside C. The complementary cycle arc, avoiding the six successive forward edges from c_{i-3} to c_{i+3}, has length $\ell - 6$, which is positive and odd. Their union is therefore a simple odd cycle of length $\ell - 2$, a contradiction. Thus the graph induced on S is a matching. Each vertex of S has at most one neighbor in S, one on C, and at least $k-2$ in $R = V \setminus (C \cup S)$. Hence there are at least $\ell(k-2)^2$ edges from S to R.

For $x \in R$, two neighbors in a single S_i would likewise give a quadrilateral. If x has neighbors in S_i and S_j, the external path of length four between c_i, c_j shows that $j - i \in \{\pm 2, \pm 4\}$. Thus

the set of indices of its neighbors in S is a clique in the graph joining indices at differences $\pm 2, \pm 4$. Multiplication by the inverse of two modulo the odd integer ℓ identifies this graph with the square of the ℓ-cycle. Its clique number is three for $\ell \geq 7$. For clarity, in a clique containing 0, the other indices lie in $\{-2, -1, 1, 2\}$; the pairs $(-2, 1), (-1, 2), (-2, 2)$ are not edges, so at most two of those four indices can be chosen. The indices $0, 1, 2$ exhibit a clique of size three.

Each vertex of R therefore receives at most three edges from S. Adding $|C| = \ell$, $|S| = \ell(k-2)$ and the resulting lower bound on $|R|$ proves (1). The argument applies to an odd-cycle component if the graph is disconnected; additional vertices only strengthen it. □

2 Closing the source extremality argument

The edge-rooted Moore-tree argument gives, for a girth-six $\mathrm{egr}(v, k, 6, \lambda)$,

$$v \geq 2(k^2 - k + 1) + \left\lceil \frac{2((k-1)^3 - \lambda)}{k} \right\rceil . \tag{2}$$

For bipartite graphs the last term can be replaced by twice the ceiling with numerator $(k-1)^3 - \lambda$. These bounds, and the biaffine construction and its cycle count, are the exact inputs already written in the parent L025 candidate. Substitution of $k = q$, $\lambda = (q-1)^2(q-2)$ in (2) gives $v \geq 2q^2 - 1$; the bipartite bound gives $v \geq 2q^2$. The biaffine graph attains the latter order. Odd q forces even v by the handshaking identity. Here the bounds are the corrected Moore-tree formulas, not a literal transcription of the printed even-girth exponents in Theorems 4.1–4.2.

Corollary 3. *For every even prime power $q \geq 8$, no $\mathrm{egr}(2q^2 - 1, q, 6, (q-1)^2(q-2))$ exists. Consequently the biaffine graph is unrestrictedly extremal for those q, as well as for every odd prime power $q \geq 3$.*

Proof. A putative graph at the displayed odd order cannot be bipartite by the bipartite lower bound. Proposition 2 gives $3v \geq 7(q^2 - q + 1)$. But

$$7(q^2 - q + 1) - 3(2q^2 - 1) = (q-5)(q-2) > 0$$

for $q \geq 8$, a contradiction. The odd-degree conclusion was established immediately above. □

The retained exhaustive computation of Goedgebeur and Jooken, Table 2, row $(k, g, \lambda) = (4, 6, 18)$, has lower and upper bounds both equal to 32. Thus it excludes the remaining order 31 at $q = 4$. This is an external finite completeness premise, not a new execution or consequence of Proposition 2. Since an even prime power is a power of two, this is the only additional case needed. Consequently, for every prime power $q \geq 3$,

$$n\big(q, 6, (q-1)^2(q-2)\big) = 2q^2.$$

In particular the possible order 127 at $q = 8$ is excluded without a new search: every nonbipartite 8-regular graph of girth at least six has at least $7(64 - 8 + 1)/3 = 133$ vertices. The uniform shell argument alone gives only 31 at $q = 4$, so it does not replace the stated external finite premise. The source's range must retain $q \geq 3$, since the biaffine graph at $q = 2$ has girth eight rather than six.

3 Scope and attribution

This is a bounded elementary resolution of one principal recovery gap in the fixed seventeen-page preprint, physical page 13, Theorem 4.3 and its comparison with Theorems 4.1–4.2. It does not revisit the accepted Suzuki calculation, supply a new Lean certificate, add an original,

or assign a numerical factor. The source's use of a bipartite comparison bound alone is insufficient for its unrestricted claim; that observation is a proof gap, not a theorem counterexample. Rewrite_termination derived the initial shell bound and the cycle-square clique argument. Root supplied the decisive improvement that the first shell is a matching, and rewrite_termination independently checked and incorporated that argument. The written shell proof is uniform and has no finite-program premise; the degree-four case retains the named external exhaustive result. Its small optional checks are diagnostics only. No priority or novelty claim for the girth-pair bound is made.

References

[1] G. Araujo-Pardo and D. Leemans, *Edge-girth-regular graphs arising from biaffine planes and Suzuki groups*, arXiv:2108.06636v1, 15 August 2021, fixed seventeen-page edition.

[2] J. Goedgebeur and J. Jooken, *Exhaustive generation of edge-girth-regular graphs*, arXiv:2401.08271v2, 25 June 2024, Table 2.